

More Agents Isn't Always Better — A Task-Dependent Theory

① Diverse NLP Tasks

Medical QA



First-line treatment for hypertension?

K = 0.88

Legal Classification



Is this contract clause enforceable?

K = 0.80

Formal Logic



If $P \rightarrow Q$ and $\neg Q$, what is P ?

K = 0.50

Long-context QA

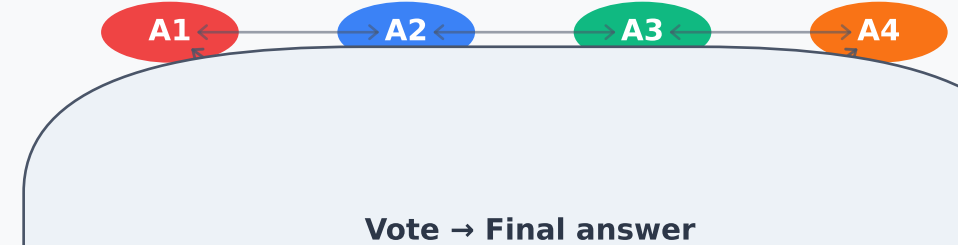


What was the CEO's main concern in Q3?

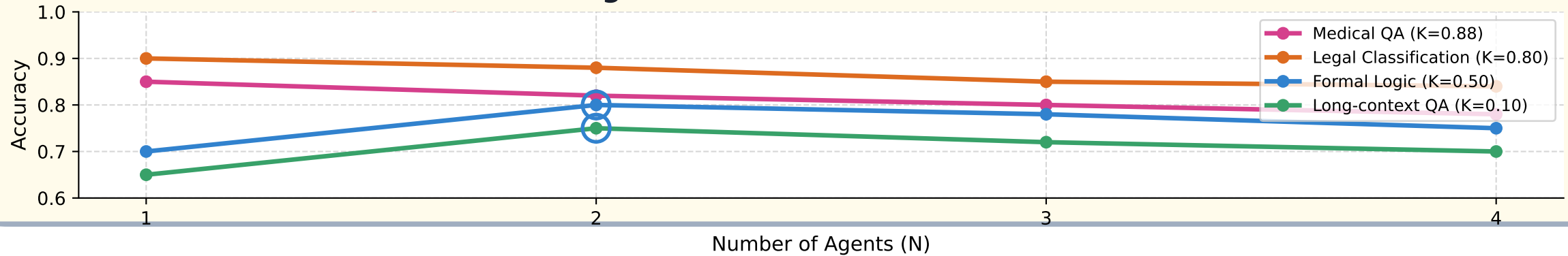
K = 0.10

② Standard Multi-Agent Protocol

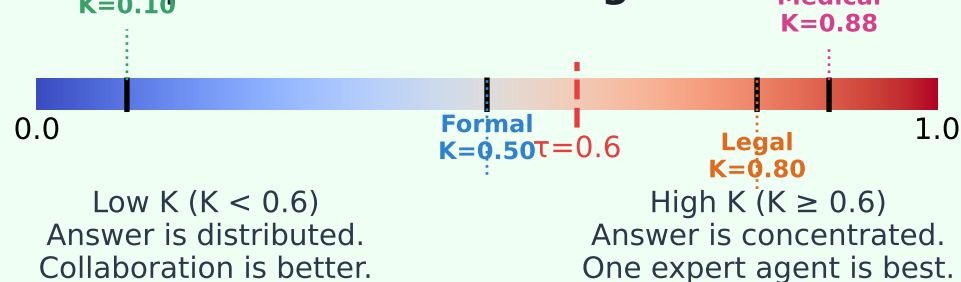
Propose, Debate



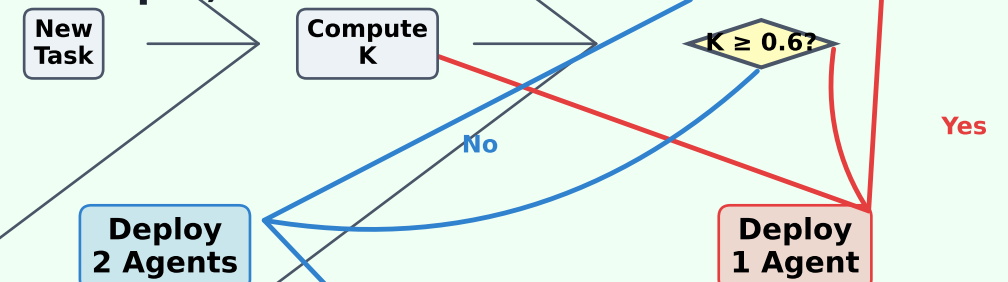
③ The Puzzle: Performance Diverges



④ The Explanation: Knowledge Concentration (K)



⑤ A Simple, Predictive Rule



Validated: 88% LOO-CV accuracy (7 of 8 trustable cells).